Systematic Mapping of Polarization-based Navigation Methods

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Outline

1. Research Context
2. Research Questions
3. Search Method
4. Preliminary Search Results
5. Challenges and Next Steps

Motivation

Currently there are only two **self-sufficient** means for computation of bearing that has been widely accepted in technical systems [1]:

- **Magnetic compass**:
  - Suffers from distortion of the magnetic field by electric currents in the airframe
  - Geomagnetic field is not uniform which requires additional compensation scheme
- **Gyrocompass**:
  - Have strong anti-interference ability but accumulate error over time [2]
  - Relative position according to the initial position instead of absolute position
  - Long time to stabilize in case of a power failure

[1] A. Dawadee, Towards an optical aircraft navigation system, Applied Mechanics and Materials 629: 321-326 (2014).
[2] J. Liu et al, Global positioning method based on polarized light compass system, Review of scientific instruments 89: 054503 (2018).

A solution inspired by nature...

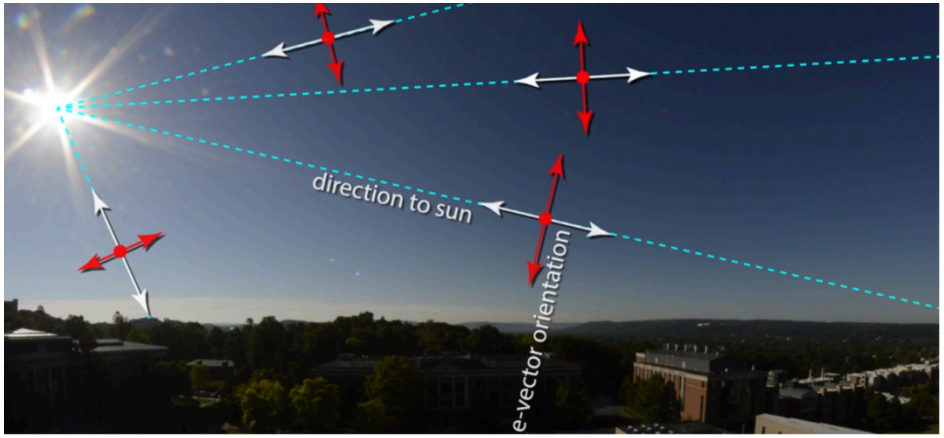
Case study: The Sahara desert ant

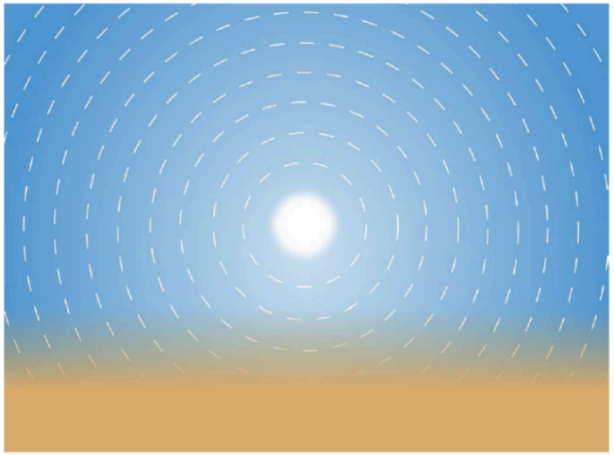
- Desert environment poses existential challenges:
  - Pheromone trails employed by other ant species for navigation cannot survive high temperature
  - No visual cues for use as path markers
  - Foraging durations are limited by intense heat
- Despite challenges, the ant manages to thrive!
- The remarkable navigation ability is achieved with the limited resources available

What is the underlying physical principle?

Polarization of Skylight

- When sunlight passes through the atmosphere, it becomes partially polarized
- Electric field for the polarized light becomes tangent to the sun's circumference
- Polarization becomes stronger with increasing distance from the sun
- The position of sun can be determined from analyzing the polarization of light





Objectives

1. Synthesize research on navigation via skylight polarization
2. Provide a reference from which to offset future contributions
3. Prepare a literature review for my master's thesis

> there are systematic reviews that ask questions that are primarily of interest to researchers. Such reviews ask questions that identify and/or scope future research activities [1]

[1] Kitchenham, Barbara. (2004). Procedures for Performing Systematic Reviews. Keele, UK, Keele Univ.. 33.

Prior Search Results

"polarization" AND "sky" AND "navigation" AND ("simulation" OR "model") AND ("camera" OR "sensor" OR "lens")

- Search over all fields
- Date range from 2016 through 2026
- Test results for control group of known good publications

Database	Number of Publications
ACM	10
Scopus	779
Web of Science	35
IEEEXplore	19
Total	843

All results retrieved on Feb. 23, 2026

Changes from Proposal

- Expand objective to *full* mapping study of polarization navigation
- Limit prior objective (investigation of camera models) to one research question
- Mapping study rather than review

Motivation: I need to prepare a *full* mapping study for my thesis, so why should I limit the objectives to just camera models in this case?

Research Questions: Overview

Methodological Questions:

- What are the dominant sensor technologies used to measure skylight polarization?
- What computational methods are used to extract navigational cues from skylight polarization?

Performance and Evaluation Questions:

- What evaluation methods are used to quantify the performance of skylight polarization navigation systems?
- How has the performance of skylight polarization navigation systems changed over time?

Search Method: Procedure Overview

1. Search string generation and piloting (iterate)
2. Develop exclusion criteria
3. Title and abstract screening
4. Develop data extraction form
5. Full-text screening and data extraction

Search Method: Search String Generation

Concept	Keywords
Skylight Polarization	skylight polarization, sky polarization, polarized skylight, polarization pattern
Navigation	navigation, orientation, heading estimation, localization, positioning, compass
Autonomous Systems*	robot, robotic, UAV, drone, car, autonomous vehicle, navigation system

- Balancing recall against precision
- Alternative spellings e.g., localisation
- AND together concepts; OR together keywords
- *Include this concept to avoid biology-only publications

Search Method: Piloting with Known Relevant Papers

Determine strength of search string by ensuring known relevant papers are included.

- Build list of relevant papers from preliminary literature review
- Include my previous publications in this list

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RESEARCH ARTICLE

Bio-Inspired Polarization Compass for Solar Azimuth Prediction Under Clear and Cloudy Sky Conditions

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Search Method: Publication Collection

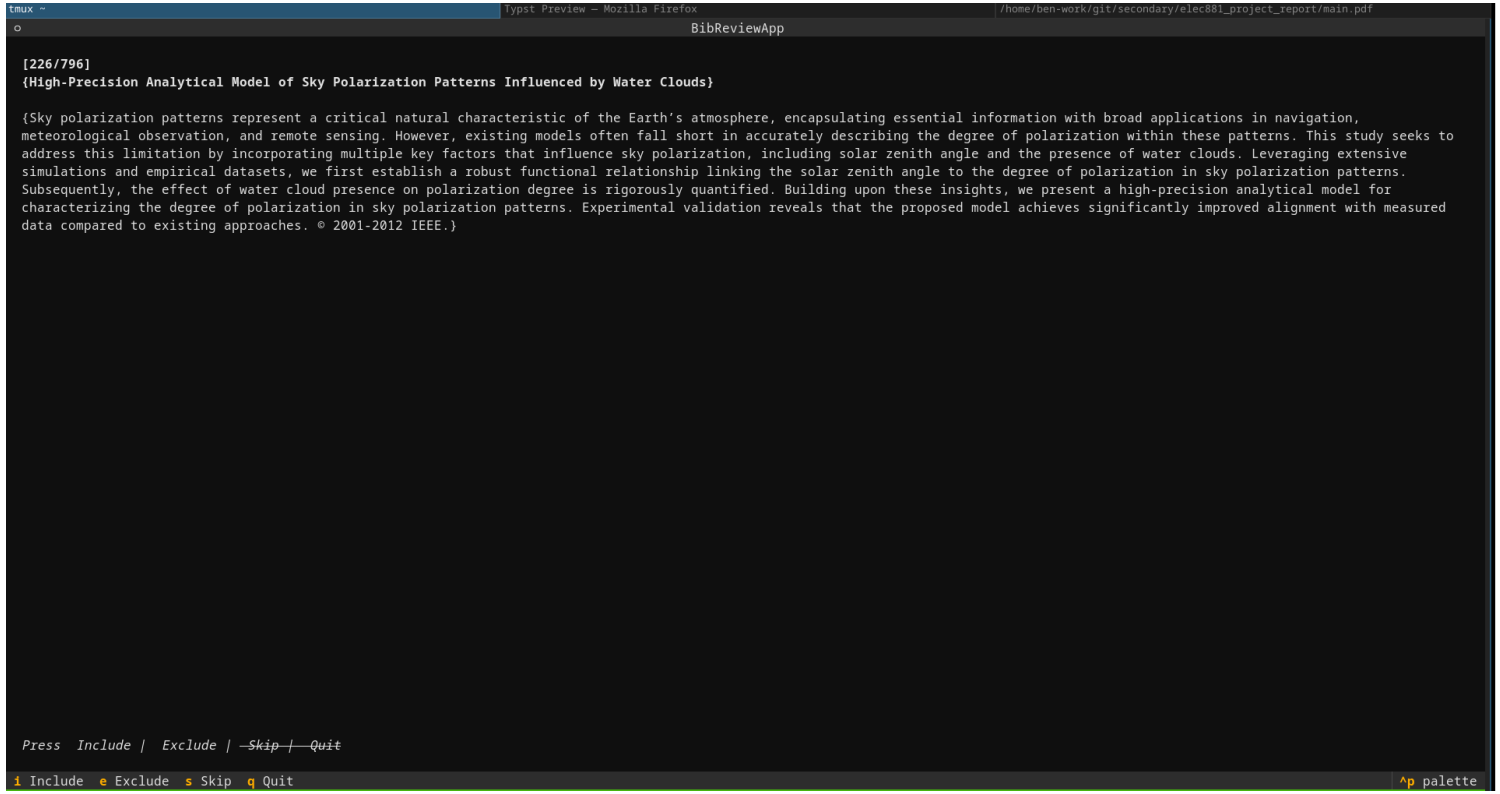
1. Download results from databases in RIS format
   - Scopus, IEEExplore, Web of Science, ACM
   - Title, author, abstract, venue, date, etc.
2. Combine results and deduplicate
   - Deduplication by unique DOI
3. Assign unique citation key to each entry

Search Method: Exclusion Criteria

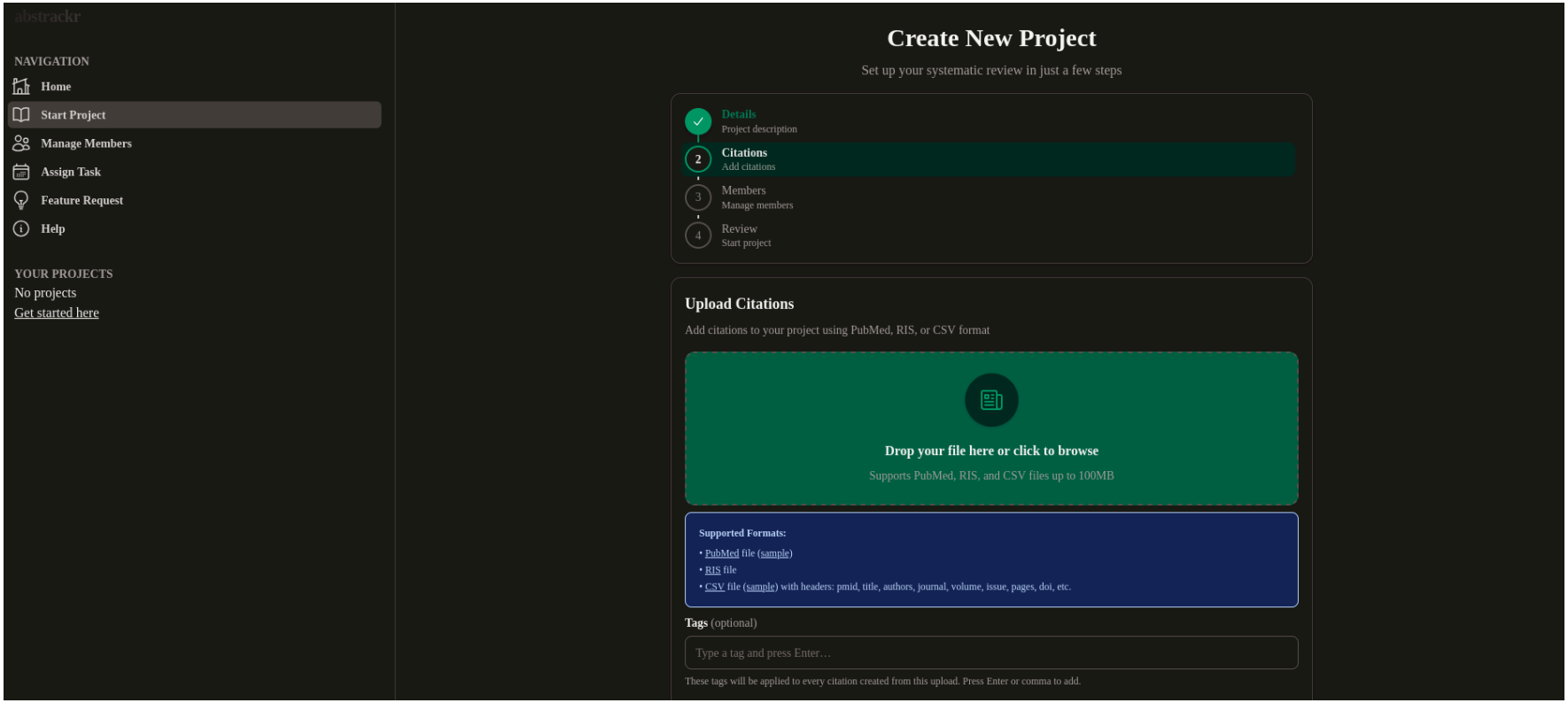
1. English language title and abstract are not available
2. DOI is not available (or does not exist)
3. No contribution toward application of skylight polarization to navigation
4. Publication is not a primary study

Search Method: Title and Abstract Screening

Custom Bib Review Tool



Abstrackr Review Tool



Search Method: Classification Scheme

Category	Fields
Bibliographic Information	Paper ID & Title, Authors & Year, Venue & Publisher
Application Context	Platform (robot / UAV / vehicle), Environment (outdoor / indoor / sim), Navigation task, Bio-inspired approach (Y/N)
Sensor Technology (RQ1)	Sensor type & category, Polarization params (AoP, DoP, Stokes), Spectral range & channels, Commercial vs. custom
Computational Methods (RQ2)	Algorithm category, Heading extraction method, Sky model (e.g., Rayleigh), Sensor fusion & real-time capability
Evaluation Methodology (RQ3)	Evaluation environment, Experiment type, Ground truth source, Dataset availability & # trials
Performance Metrics (RQ4)	Heading & max error (°), Localization error, Robustness evaluation, Computational cost

Challenges and Next Steps

Challenges:

- Defining research questions and scope
- Balancing recall against precision in the search string
- Managing large numbers of publications efficiently

Next steps:

- Apply new search string to databases
- Complete title and abstract screening
- Refine and apply data extraction form during full-text screening